

Object Oriented Programming

Lab Assessment 1 (10 Marks)

Task-1 (Click [here](#) to open the compiler) (3 Marks)

Write a program that computes the value of e^x by using the formula.

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Prompt the user for the desired accuracy of e (i.e., the number of terms in the summation).

Task-2 (Click [here](#) to open the compiler) (3 Marks)

Write a program that calculates GCD of two numbers.

$$\begin{aligned} 36 &= 2 \times 2 \times 3 \times 3 \\ 60 &= 2 \times 2 \times 3 \times 5 \\ \text{GCD} &= \text{Multiplication of Common Factors} \\ &= 2 \times 2 \times 3 \\ &= 12 \end{aligned}$$

Task-3 (Click [here](#) to open the compiler) (4 Marks)

A mobile phone service provider has three different subscription packages for its customers:

Package A: For \$39.99 per month 450 minutes are provided. Additional minutes are \$0.45 per minute.

Package B: For \$59.99 per month 900 minutes are provided. Additional minutes are \$0.40 per minute.

Package C: For \$69.99 per month unlimited minutes provided.

Write a program that calculates a customer's monthly bill. It should ask which package the customer has purchased and how many minutes were used. It should then display the total amount due.

Input Validation: Be sure the user only selects package A, B, or C.